1. Differential Equations

- Form a differential equation using the information given
  - If something is proportional, add constant of proportionality k
  - If rate is decreasing, add a negative sign
- Separate variables, bring dx and dt on opposite sides
  - Integrate both sides to form an equation
  - Add arbitrary constant
  - Use conditions given to find c and/or k

{W10-P33} Question 9:

A biologist is investigating the spread of a weed in a particular region. At time t weeks, the area covered by the weed is Am^2 . The biologist claims that rate of increase of A is proportional to $\sqrt{2\mathrm{A}-5}$.

1. Write down a differential equation given info
2. At start of investigation, area covered by weed was $7m^2$. 10 weeks later, area covered $= 27m^2$ Find the area covered 20 weeks after the start of the investigation.

Solution:

Part (i)

$$\frac{\mathrm{dA}}{\mathrm{dt}} \propto \sqrt{2A-5} = k\sqrt{2A-5}$$

Part (ii)

Proceed to form an equation in A *and* t *:*

$$\frac{\mathrm{dA}}{\mathrm{dt}} = k\sqrt{2A-5}$$

Separate variables

$$\frac{1}{\sqrt{2A-5}}dA = kdt$$

Integrate both sides

$$kt + c = (2A-5)^{\frac{1}{2}}$$

When $t = 0$ *:*

$A = 7 \therefore c = 3$

$$kt + 3 = (2A-5)^{\frac{1}{2}}$$

When $t = 10$ *:*

$$10k + 3 = (2(27)-5)^{\frac{1}{2}}$$

$$10k = \sqrt{49} - 3$$

$$k = 0.4$$

Now substitute 20 *as* t *and then find* A *:*

$$0.4(20) + 3 = (2A-5)^{\frac{1}{2}}$$

$$11 = (2A-5)^{\frac{1}{2}}$$

$$121 = 2A - 5$$

$$A = 63m^2$$

{S13-P31} Question 10:

Liquid is flowing into a small tank which has a leak. Initially the tank is empty and, t minutes later, the volume of liquid in the tank is $\mathrm{V}\ cm^3$. The liquid is flowing into the tank at a constant rate of $80\ cm^3$ per minute. Because of the leak, liquid is being lost from the tank at a rate which, at any instant, is equal to $\mathrm{kV}\ cm^3$ per minute where k is a positive constant.

i. Write down a differential equation describing this situation and solve it to show that:

$$V = \frac{1}{k}\left(80 - 80e^{-kt}\right)$$

$$k = \frac{4 - 4e^{-15k}}{25}$$

ii. Work out the volume of liquid at $t = 20$ and state what happens to volume after a long time.

Solution:

Part (i)

Represent the given information as a derivative:

$$\frac{\mathrm{dV}}{\mathrm{dt}} = 80 - kV$$

Proceed to solve the differential equation:

$$\frac{\mathrm{dt}}{\mathrm{dV}} = \frac{1}{80 - kV}$$

$$dt = \frac{1}{80 - kV}\mathrm{dV}$$

$$\int (1)\,\mathrm{dt} = \int \frac{1}{80 - kV}\mathrm{dV}$$

$$t + c = -\frac{1}{k}\ln|80 - kV|$$

Use the given information; when $t = 0,\ V = 0$ *:*

$$\therefore c = -\frac{1}{k}\ln(80)$$

Substitute back into equation:

$$t - \frac{1}{k}\ln(80) = -\frac{1}{k}\ln|80 - kV|$$

$$t = \frac{1}{k}\ln(80) - \frac{1}{k}\ln|80 - kV|$$

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$$t = \frac{1}{k}\ln\left(\frac{80}{80-kV}\right)$$

$$kt = \ln\left(\frac{80}{80-kV}\right)$$

$$e^{\mathrm{kt}} = \frac{80}{80-kV}$$

$$80 - kV = \frac{80}{e^{\mathrm{kt}}}$$

$$kV = 80 - 80e^{-kt}$$

$$V = \frac{1}{k}\left(80 - 80e^{-kt}\right)$$

Part (ii)

After carrying out the iterations, the following result will be obtained:

$$k = 0.14\ \ (2d.p.)$$

Part (iii)

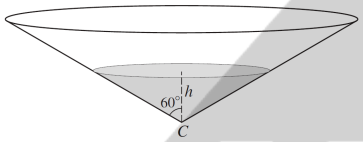
Simply substitute into the equation's t*:*

$$V = \frac{1}{0.14}\left(80 - 80e^{-0.14(20)}\right) = 537\ cm^3$$

The volume of liquid in the tank after a long time ***approaches*** *the max volume:*

$$V = \frac{1}{0.14}(80) = 571\ cm^3$$

{W13-P31} Question 10:



A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is 60°, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time *t*. The tank becomes empty when $t = 60$.

$$\frac{\mathrm{dh}}{\mathrm{dt}} = -Ah^{-\frac{3}{2}}$$

i. Solve differential equation given in part i and obtain an expression for t in terms of h and H.

Solution:

Part (i)

First represent info they give us as an equation:

$$V = \frac{1}{3}\pi r^2 h$$

$$r = \tan 60 \times h = h\sqrt{3}$$

$$\therefore V = \frac{1}{3}\pi\left(h\sqrt{3}\right)^2 h = \pi h^3$$

$$\frac{\mathrm{dV}}{\mathrm{dh}} = 3\pi h^2$$

$$\frac{\mathrm{dV}}{\mathrm{dt}} \propto -\sqrt{h} = -kh^{\frac{1}{2}}$$

Find the rate of change of h*:*

$$\frac{\mathrm{dh}}{\mathrm{dt}} = \frac{\mathrm{dV}}{\mathrm{dt}} \div \frac{\mathrm{dV}}{\mathrm{dh}}$$

$$\frac{\mathrm{dh}}{\mathrm{dt}} = \frac{-kh^{\frac{1}{2}}}{3\pi h^2} = -\frac{k}{3\pi}h^{-\frac{3}{2}}$$

Part (ii)

$$dt = \frac{1}{-Ah^{-\frac{3}{2}}}\mathrm{dh}$$

$$\int \mathrm{Adt} = \int \frac{1}{-h^{-\frac{3}{2}}}\mathrm{dh}$$

$$At + c = -\frac{2}{5}h^{\frac{5}{2}}$$

Use given information to find unknowns; when $t = 0$ *:*

$$-A(0) + c = \frac{2}{5}(H)^{\frac{5}{2}} \quad \therefore c = \frac{2}{5}H^{\frac{5}{2}}$$

When $t = 60$ *:*

$$-A(60) + c = 0$$

$$c = 60A$$

$$A = \frac{1}{150}H^{\frac{5}{2}}$$

Thus the initial equation becomes:

$$-\frac{1}{150}H^{\frac{5}{2}}t + \frac{2}{5}H^{\frac{5}{2}} = \frac{2}{5}h^{\frac{5}{2}}$$

$$H^{\frac{5}{2}}\left(-\frac{t}{150} + \frac{2}{5}\right) = \frac{2}{5}h^{\frac{5}{2}}$$

$$-\frac{t}{150} + \frac{2}{5} = \frac{2h^{\frac{5}{2}}}{5H^{\frac{5}{2}}}$$

$$\frac{t}{150} = \frac{2}{5} - \frac{2h^{\frac{5}{2}}}{5H^{\frac{5}{2}}}$$

$$t = 150\left(\frac{2}{5} - \frac{2h^{\frac{5}{2}}}{5H^{\frac{5}{2}}}\right) = 60 - 60h^{\frac{5}{2}}H^{-\frac{5}{2}}$$

$$t = 60\left(1 - \left(\frac{h}{H}\right)^{\frac{5}{2}}\right)$$

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